Separating Safety from Preference in Clinical LLM Evaluation

Keep clinical safety visible when optimizing for preference

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When preference hides risk

Patients ask rehabilitation agents about pain, fatigue, and exercise progression. A fluent, empathetic answer can still give clinically inappropriate guidance.

Record clinical safety beside preference. Reviewers can then audit a response even when raters prefer it.



Figure A: mRehab's smartphone sits in a 3D-printed exercise object; its sensors track repetitions and movement quality. The mRehab Advisory Agent answers patient questions using exercise data and clinical knowledge.

Separate the judgments

- Six domain experts score safety (yes/no) for harm or clinical irresponsibility, and correctness (1–5) for clinical facts and interpretation.
- Three end users score helpfulness (1–5) for practical rehabilitation use, and affect (1–5) for an empathetic, encouraging, and motivational tone.
- Both cohorts score clarity (1–5) for concise, coherent answers on a mobile screen, then rank three blinded responses best to worst, with no ties.

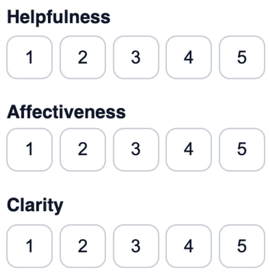


Figure B: End-user rating controls, cropped from the scoring screen. The screen's "Affectiveness" is the affect score.

Blinded collection

- Each fixed bundle contains a question and three candidate responses.
- A stable hashed order maps conditions to A/B/C for each rater and question.
- Condition identities stay server-side until submission.
- Raters score each response, then rank all three; the backend validates the order.

Raters received no calibration trial. Of 200 bundles, 39 were rated by both cohorts.

ROLES BEHIND THE SCREEN

The platform separates tester, rater, and admin workflows. This pilot uses its blinded rating instrument. Rubrics assign clinical judgments to experts and practical use to end users.

KEEP THE RECORD INTACT

Store response text, rank, cohort, and rubric values together. Export three scored audit rows, one preferred-response training example, or chosen–rejected preference pairs.

A rank-only export loses safety labels. Keeping the full record lets reviewers trace disagreement to a rater or question.

A rehabilitation pilot

435
ranked triples

1,305
scored responses

Nine raters evaluated a 200-question pool. The 435 judgments comprise 314 expert and 121 end-user evaluations; each scores three candidates.

Judgments cover General Knowledge (182), Trend (94), Advice (60), and Summary (99).

SAME BACKBONE, DIFFERENT CONTEXT

All conditions use GPT-4.1 mini. Architecture, data access, and clinical layering change together.

Condition	Context and tools
mRehab	Orchestrator with Data Analysis, Health Expert, and Rehab Coaching subagents.
Tool-GPT	Python tool retrieves the patient's exercise data.
Text-GPT	Full patient data in a markdown table; no tools.

Preference differs from helpfulness

53.3%
mRehab ranked first: 232/435 judgments

Tool-GPT ranked first in 133/435 judgments (30.6%); Text-GPT in 70/435 (16.1%). Text-GPT scored highest on user helpfulness, **4.01 vs. 3.91** for mRehab.

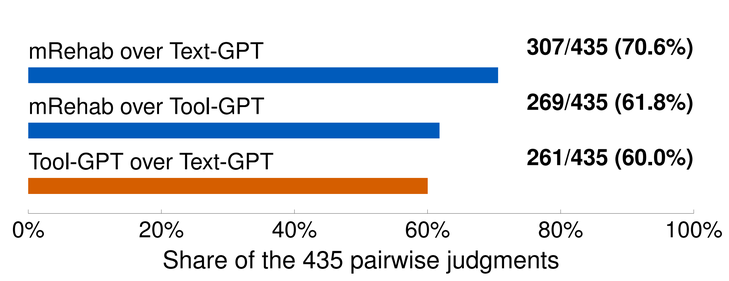


Figure C: Pairwise preference across all 435 rankings: mRehab wins 70.6% against Text-GPT and 61.8% against Tool-GPT.

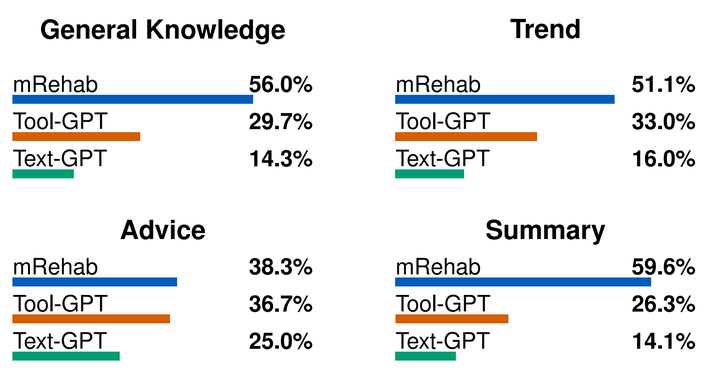


Figure D: First-place share by category. mRehab leads in General Knowledge, Trend, and Summary; Advice is nearly tied.

SCORES BY CONDITION

Measure	mRehab	Tool-GPT	Text-GPT
Expert safe rate	88.8%	93.7%	93.7%
Expert correctness	4.31	4.21	4.18
Expert clarity	4.38	3.99	3.02
User helpfulness	3.91	3.73	4.01
User affect	4.08	3.12	3.23
User clarity	4.27	3.96	3.37

mRehab leads on affect and clarity. Expert correctness is similar across conditions (4.18–4.31).

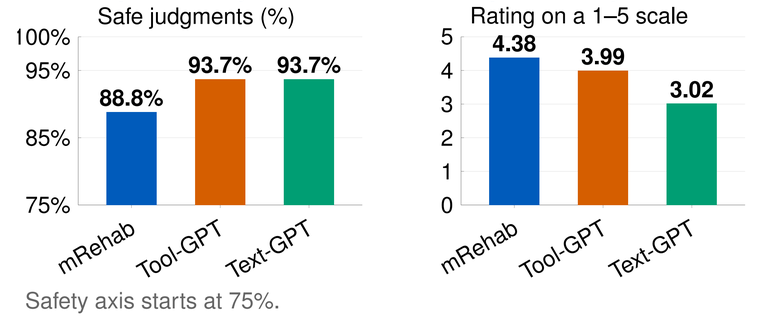
Advice calls for caution and stopping rules.

Comparisons are exploratory, with rater-clustered estimates and no correction across the full multiple-test family.

Safety needs its own label

88.8%
mRehab expert safety yes-rate

Each comparator reaches **93.7%**. Question-level means do not establish clinical outcomes.



Safety axis starts at 75%.

Figure E: Expert safety and clarity by condition. mRehab has lower aggregate safety and higher clarity.

Check who disagrees

29 of 36
mRehab unsafe labels came from one expert

Raw safety: 278/314 safe, so 36 unsafe. The other five experts flagged seven. Only one mRehab question drew unsafe labels from multiple experts.

EXCLUDING THAT EXPERT

Condition	Safe rate
mRehab	97.4%
Tool-GPT	96.6%
Text-GPT	97.6%

Expert safety agreement is near zero: **Krippendorff's α = 0.05**. Strictness versus miscalibration remains unresolved.

AGREEMENT LIMITS THE CLAIM

Agreement is also low for other expert scores:

- Correctness: α = 0.12.
- Clarity: α = 0.26.
- Advice has the lowest agreement across categories, α = –0.08.

For safety, α = –0.07 for mRehab, 0.09 for Tool-GPT, and 0.23 for Text-GPT. Experts disagree most about the preferred agent's safety.

The safety gap remains unresolved. Rater-cluster bootstrap uncertainty includes no difference. Rater provenance locates the concentration in one expert.

Design for clinical auditing

- Keep safety labels alongside preference throughout alignment.
- Calibrate raters and balance assignments before collection.
- Collect written rationales for low scores.
- Expand Advice evaluation and test multi-turn safety.

LIMITS ON INTERPRETATION

Nine raters gave 1–60 judgments each. Conditions co-vary, so causal claims need controlled ablations.

Keep both labels for clinical auditing.

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